QQ-M-55C
December 1, 1970
SUPERSEDING
Fed. Spec. QQ-M-55b
September 5, 1962

FEDERAL SPECIFICATION

MAGNESIUM ALLOY, PERMANENT AND SEMI-PERMANENT MOLD CASTINGS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification presents requirements for magnesium alloy permanent and semi-permanent mold castings.

1.2 Classification.

1.2.1 Compositions and tempers. Magnesium alloy permanent and semi-permanent mold castings covered by this specification shall be of the following compositions and tempers, as specified (see 6.2):

Compositions:

AM100A	AZ91C	HK31A
AZ63A	AZ92A	HZ32A
AZ81A	EZ33A	QE22A

Tempers:

F - As fabricated.
T4 - Solution heat treated.
T5 - Artificially aged.
T6 - Solution heat treated and artificially aged.
T61 - Solution heat treated and artificially aged and straightened.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC MECA

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Specifications:

MIL-M-3171 - Magnesium Alloy, Process for Pretreatment and Prevention of Corrosion.
MIL-C-6021 - Casting, Classification and Inspection of.
MIL-M-6857 - Magnesium Alloy Castings, Heat Treatment of.
MIL-I-6869 - Impregnants for Aluminum Alloy and Magnesium Alloy Castings.
MIL-W-18326 - Welding of Magnesium Alloys, Gas and Arc, Manual and Machine Processes for.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-276 - Impregnation of Porous Nonferrous Metal Castings.
MIL-STD-453 - Inspection, Radiographic.
MIL-STD-649 - Aluminum and Magnesium Products, Preparation for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material. The material shall be such as to produce permanent and semi-permanent mold castings in full compliance with the requirements specified herein.

3.2 Manufacture. The castings shall be made by the permanent or semi-permanent mold processes.

3.3 Foundry control. Unless otherwise stated, castings shall be produced under foundry control approved by the procuring activity. Foundry control shall consist of examination of castings by radiographic inspection in accordance with the requirements of MIL-STD-453 or by other approved methods for determining internal defects until the gating, pouring and other foundry practices have been established to produce castings meeting the quality standards furnished by the procuring activity or as agreed upon by the procuring activity and contractor. When foundry practices have been so established, the production method shall not be changed without demonstrating to the satisfaction of the procuring activity that the change does not adversely affect the quality of the castings.

3.4 Soundness.

3.4.1 When specified (see 6.2), the soundness of castings shall conform to standards furnished or approved by the procuring activity. The extent of porosity, cold shuts and other defects in the castings shall not exceed those shown by the standard, and such defects shall be absent in designated areas.

3.4.1.1 The standards supplied or approved by the procuring activity for determining conformance with soundness requirements may consist of sectioned castings or photographs thereof or radiographs of at least the important sections of the castings.

3.5 Identification. Where practicable and unless otherwise specified (see 6.2), each casting shall be identified with the part number by the use of raised numerals in a location indicated on the drawing. When no location is shown on the drawing, the number shall be so located as not to be machined off in finishing to the required dimensions. When practicable, each casting shall also be marked with the lot number.

3.5.1 If the location of impression markings is not specified on the drawing, it shall be fixed by the contractor preferably on a cast-in, raised pad, in such a manner as to minimize the danger of its having a deleterious effect on the static or fatigue strength of the casting. The raised pad should be located in an unmachined area of the casting, and preferably adjacent to the part number area, so that mechanical interference is avoided.

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3.5.2 Marking other than impression stamping shall be applied to the material by means of a hand stamp and a suitable marking fluid or other approved means. All marking and marking symbols such as those specified in 3.11.3.4, 3.11.4, and 3.11.5 for repaired castings shall be clearly legible.

3.6 Straightness. Castings shall be furnished in condition ready for machining and shall not require further straightening. After the yield strength of the castings has been increased by an aging operation, straightening shall not be permitted except that minor restraightening to remove any slight warping caused by aging or machining shall be permitted, if approved by the procuring activity.

3.7 Heat treatment.

3.7.1 Castings shall be heat treated in such manner as to produce material with the utmost uniformity which will conform to the properties specified herein. Heat treatment shall be performed on the whole casting and never on a portion only.

3.7.2 Requirements for the heat treatment of castings shall be in accordance with MIL-M-6857, or as approved by the procuring activity.

3.8 Chemical composition.

3.8.1 The chemical composition of the material shall be within the limits specified in table I.

3.8.2 The contractor shall furnish an analysis of each melt showing the percentage of each of the elements designated in table I.

3.8.3 Chemical analysis by the contractor of the individual melts may be waived by the procuring activity, provided that the foundry's method of composition control is acceptable to him, or that all the material in the lot can be identified as being from melts previously analyzed and found to be in conformance with the chemical composition requirements of the alloy specified.

3.9 Mechanical properties.

3.9.1 Unless otherwise specified in the contract or order, the mechanical properties as determined from separately cast test bars shall conform to the requirements shown in table II.

3.9.2 Specimens cut from castings. Under any of the following conditions, conformance to mechanical property requirements shall be determined by tests on specimens cut from castings:

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TABLE I. Chemical requirements 1/

Composition	Magnesium	Aluminum	Zinc	Manganese, minimum	Silicon, maximum	Copper, maximum	Nickel, maximum	Silver	Rare earth	Zirconium	Thorium	Total other elements maximum
	percent	percent	percent	percent	percent	percent	percent	percent	percent	percent	percent	percent
AM100A	Remainder	9.3-10.7	0.30 max.	0.10	0.30	0.10	0.01					0.30
AZ63A	Remainder	5.3-6.7	2.5-3.5	0.15	0.30	0.10	.01					.30
AZ81A	Remainder	7.0-8.1	0.40-1.0	.13	0.30	0.10	.01					0.30
AZ91C	Remainder	8.1-9.3	0.40-1.0	0.13	.30	.10	.01					0.30
AZ92A	Remainder	8.3-9.7	1.6-2.4	0.10	.30	.10	.01					.30
EZ33A	Remainder		2.0-3.1			.10	.01		2.5-4.0	0.50-1.0		.30
HK31A	Remainder		0.30 max.			.10	.01			0.50-1.0	2.5-4.0	.30
HZ32A	Remainder		1.7-2.5			.10	.01		0.10 max.	0.50-1.0	2.5-4.0	0.30
QE22A	Remainder					.10	.01	2.0-3.0	1.8-2.5 2/	0.40-1.0		0.30

1/ Analysis shall regularly be made only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that these other elements are not present in excess of the above limits.

2/ Didymium, cerium-free rare earth metals consisting of a mixture primarily of praseodymium (Pr) and neodymium (Nd).

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TABLE II. Mechanical property requirements for separately cast specimens

Alloy and temper	Tensile strength, minimum	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 inches, minimum
		Minimum	Extension under load	
	p.s.i.	p.s.i.	inches per inch	percent
AM100A				
F	20,000	10,000	0.0035	-
T4	34,000	10,000	.0035	6
T6	34,000	15,000	.0043	2
T61	34,000	17,000	.0046	-
AZ63A				
F	26,000	11,000	.0037	4
T4	34,000	11,000	.0037	7
T5	26,000	12,000	.0038	2
T6	34,000	16,000	.0045	3
AZ81A				
T4	34,000	11,000	.0037	7
AZ91C				
F	23,000	11,000	.0037	-
T4	34,000	11,000	.0037	7
T5	23,000	12,000	.0038	2
T6	34,000	16,000	.0045	3
AZ92A				
F	23,000	11,000	.0037	-
T4	34,000	11,000	.0037	6
T5	23,000	12,000	.0038	-
T6	34,000	18,000	.0048	-
EZ33A				
T5 Room Temp.	20,000	14,000	.0042	2
T5 500° F	13,000	8,000	-	-
HK31A				
T6 Room Temp.	27,000	13,000	.0040	4
T6 500° F	21,000	13,000	-	-
HZ32A				
T5 Room Temp.	27,000	13,000	.0040	4
T5 500° F	13,000	8,000	-	-
QE22A				
T6 Room Temp.	35,000	25,000	.0062	2
T6 600° F	13,000	10,000	.0035	-

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a. When specified on the applicable drawings.
b. When authorized by the procuring activity.
c. When separately cast specimens are not available.

The average and minimum tensile strength, tensile yield strength and elongation of test specimens cut from castings shall conform to the requirements shown in table III. See note 2/, table III.

TABLE III. Mechanical property requirements for specimens cut from castings 1/

Alloy and temper	Tensile strength		Yield strength at 0.2 percent offset or at extension indicated			
	Average	Minimum 2/	Average	Minimum 2/	Extension under load	
					For average	For minimum
	p.s.i.	p.s.i.	p.s.i.	p.s.i.	inches per inch	inches per inch
AM100A						
T4	25,500	17,000	10,000	9,000	0.0035	0.0034
T6	25,500	17,000	13,500	10,000	.0041	.0034
T61	25,500	17,000	15,500	11,500	.0044	.0038
AZ63A						
T4	25,500	17,000	10,000	9,000	.0035	.0034
T6	25,500	17,000	14,000	12,000	.0042	.0039
AZ81A						
T4	25,500	17,000	10,000	9,000	.0035	.0034
AZ91C						
T4	25,500	17,000	10,000	9,000	.0035	.0034
T6	25,500	17,000	14,500	12,000	.0042	.0039
AZ92A						
T4	25,500	17,000	10,000	9,000	.0035	.0034
T6	25,500	17,000	16,000	13,500	.0045	.0041
EZ33A						
T5 Room Temp.	15,000	13,000	12,500	11,000	.0039	.0037
T5 500° F	-	10,000	-	6,000	-	-
HK31A						
T6 Room Temp.	23,000	19,000	11,700	10,500	.0038	.0036
T6 500° F	-	14,000	-	10,500	-	-
HZ32A						
T5 Room Temp.	23,000	19,000	11,700	10,500	.0038	.0036
T5 500° F	-	10,000	-	6,000	-	-
QE22A						
T6 Room Temp.	32,000	28,000	23,000	20,000	.0056	.0051
T6 600° F	-	10,000	-	8,000	-	-

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1/ The average percent elongation of specimens cut from castings shall be not less than 25 percent of the values specified for separately cast test bars.
2/ The above minimum values are representative of the weakest areas in any type casting for the composition listed (i. e., adjacent to gates and risers,). Higher design properties for designated areas in castings are normally obtained and shall be specified in the contract drawings or order. (See 3. 9. 3.)

3. 9. 3 Mecnanical properties of designated areas of castings. When specified on the applicable drawings or in the contract or order, the mechanical properties of test specimens cut from the areas of the casting designated on the drawing or in the contract or order, shall meet the values specified on the drawing or in the contract or order. See note 2/, table III.

3. 9. 4 Mechanical properties at 500° F. Alloys EZ33A, HK31A, and HZ32A shall be capable of meeting the requirements shown in tables II and III when tested in the following manner. Tensile test specimens shall be heated to 500 ±5° F, held for ten minutes before testing and loaded at this temperature at a rate not greater than 0. 005 inch per inch per minute up to the yield strength and at a rate of 0. 10 inch per inch per minute above the yield strength.

3. 9. 4. 1 Mechanical properties at 600° F. Alloy QE22A shall be capable of meeting the requirements shown in tables II and III when tested in the following manner. Tensile test specimens shall be heated to 600 ±5° F, held for ten minutes before testing and loaded at this temperature at a rate not greater than 0. 005 inch per inch per minute up to the yield strength and at a rate of 0. 10 inch per inch per minute above the yield strength.

3. 9. 5 Simulated service tests. When the castings have critical functioning requirements, special tests to simulate the stress conditions incurred in service may be required. The requirements and methods of making the test shall be as specified by the procuring activity or as agreed upon between the procuring activity and contractor.

3. 10 Protective treatment. As a final process, all castings shall be subjected to a chrome pickle surface treatment or a dichromate surface treatment. Unless otherwise specified (see 6. 2), castings shall be given a chrome pickle treatment in accordance with type I of MIL-M-3171.

3. 11 Workmanship.

3. 11. 1 Castings shall be of uniform quality and condition, free from injurious blowholes, porosity, hard spots, shrinkage cracks, and other injurious defects and shall be smooth and cleaned before inspection.

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3.11.2 Repair of castings. Castings shall not be repaired by welding, impregnation, peening, plugging or other methods without permission of the procuring activity. Such permission may be given only when the repaired defect will not adversely affect the strength or machinability or otherwise impair the suitability of the castings for the purpose intended. Soldering shall not be performed under any conditions. Repaired castings shall be reinspected in accordance with the applicable portions of the pertinent specifications, drawings and directives.

3.11.3 Welding. When welding is permitted, it shall be done by methods suitable for the particular alloy and only in those portions of castings not highly stressed in service. Such areas shall be as indicated on the applicable drawing. Welding methods shall be in accordance with such specifications as are referenced on the drawings, as are required by the contract or order, or as specified when permission to weld is granted. Periodic radiographic checks of welded castings shall be made to ensure adherence to a satisfactory procedure for welding. Welding shall be in accordance with MIL-W-18326 except in lieu of paragraph 3.1 thereof, the welding operator shall be capable of repair welding in accordance with radiographic standards applicable to welded areas.

3.11.3.1 When castings are to be supplied in the heat-treated condition, they shall be heat treated or reheat treated, with "proper precautions against germination", to the required temper after welding, except that small arc welds may be performed without subsequent heat treatment upon approval of the procuring activity.

3.11.3.2 Unless otherwise specified (see 6.2), castings which have been repaired by welding shall be inspected radiographically after all reworking and heat treatment have been completed.

3.11.3.3 All welds shall be of high quality and free from cracks, oxide, porosity, and lack of fusion.

3.11.3.4 Welded castings shall be marked with a symbol of three concentric circles with a letter or number designating the welder adjacent to the symbol. The outer circle of the symbol shall be not larger than 1/4 inch outside diameter. All welded areas shall be encircled with a ring of white paint prior to submission for final inspection.

3.11.4 Impregnation. When impregnation is permitted, it shall be only to correct general seepage leaks and shall not be used to correct poor foundry technique or significant porosity. It shall be accomplished in accordance with such specifications as may be shown on the drawing or that may be required by the contract or order. If no specifications are stipulated, the method used shall be approved by the procuring activity. Unless otherwise authorized, castings which have been impregnated shall be marked "IMP". Impregnation of castings shall be in accordance with MIL-STD-276. Impregnants used shall be in accordance with MIL-I-6869.

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3.11.5 Peening. Where peening is permitted, it shall be only to correct localized minor seepage leaks and small surface imperfections, or to disclose subsurface voids for the purpose of inspection. Peening shall not be permitted to repair cracks, cold shuts, shrinks, mis-runs, defects due to careless handling or other similar major defects. Peening may be accomplished either hot or cold and shall be performed by methods which are acceptable to the procuring activity. Peened castings shall be marked with a maltese cross approximately 1/4 inch high.

3.11.6 Blending. Blending with suitable grinders or other tools shall be permitted only for the removal of surface imperfections and shall not result in dimensions below minimum drawing tolerances.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of not more than 1,000 pounds of cleaned castings of the same composition and temper, except where this weight limit may be exceeded by a single casting. In such cases, each casting shall be considered a lot. Unless otherwise specified in the contract or order, each lot of castings shall be from the same melt, or such groups of melts as may be designated by the procuring activity.

4.2.2 Samples for chemical and spectrographic analysis.

4.2.2.1 Sample for chemical analysis. When samples for chemical analysis are specified in the contract or order, at least one sample shall be taken from each heat or lot of castings. Samples for chemical analysis shall conform to the requirements of method 111.2 of Fed. Test Method Std. No. 151. Each sample shall consist of not less than two ounces of chips.

4.2.2.2 Samples for spectrographic analysis. When samples for spectrographic analysis are specified in the contract order, at least one sample shall be taken from each heat or lot of castings. Samples for spectrographic analysis shall conform to the requirements of method 112.2 of Fed. Test Method Std. No. 151. Each sample shall weigh approximately 20 grams.

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4.2.3 Samples for tension test.

4.2.3.1 Two or more test bars shall be cast from a melt in the lot to represent each lot. No separately cast test bars are required if conformance with mechanical property requirements is demonstrated by tests of specimens cut from castings in accordance with the requirements of 3.9.2.

4.2.3.2 Except in the case where castings are to be shipped in the "as cast" temper, the test bars from each lot shall be heat treated to the same temper with production castings of the same alloy. When test bars are used, one of the bars representative of the particular lot shall be tested and the lot judged by the result of this test.

4.2.3.3 Each heat-treating furnace charge shall be represented by at least two test bars of the same composition range. The heat treatment of a furnace charge shall be judged by these test bars. All separately cast bars shall conform to the minimum values specified in table II.

4.2.3.4 When test specimens taken from castings are required in accordance with the requirements of 3.9.2, one or more castings in the end use heat treat temper to be furnished shall be selected from each lot and not less than three tension test specimens shall be machined therefrom, of which one shall be from the thinnest section, one from the heaviest section, and one from an intermediate section. The mechanical properties of specimens so tested shall conform with the requirements of table III or shall meet the values specified on the drawings or in the contract or order.

4.2.3.5 When test specimens taken from designated areas of castings are required in accordance with the requirements of 3.9.3, one or more castings in the end use heat treat temper to be furnished shall be selected from each lot or group of lots, as may be designated. Not less than two tension test specimens shall be machined therefrom from each designated area and tested. The mechanical properties of specimens so treated shall conform with the requirements of table III or shall meet the values specified on the drawing or in the contract or order.

4.2.3.6 Tensile test specimens cut from castings shall be flat or round, conforming to the requirements of method ASTM E-8, figure 6 or 8, referenced in Fed. Test Method Std. No. 151.

4.2.3.7 Unless otherwise specified (see 6.2), test bars shall be cast in a chill mold of such design as to produce a test bar similar to the requirements of method ASTM-E8, figure 18, referenced in Fed. Test Method Std. No. 151.

4.2.3.8 The metal for the separately cast test bars shall be part of the melt used for casting. In the event the metal for castings is given any treatment such

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as fluxing, or cooling and reheating, the metal for the test bars shall be given the same treatment.

4.2.3.9 Unless otherwise specified (see 6.2), separately cast test bars shall be tested without machining except as it may be necessary to adapt them to the grips of the testing machine.

4.3 Examination and tests.

4.3.1 Soundness. Unless otherwise specified (see 6.2), when soundness is required in accordance with 3.4, it shall be determined by radiographic examination of all castings in the lot. For aeronautical application, inspection of castings shall be in accordance with MIL-C-6021 for the class assigned. For nonaeronautical use, radiographic inspection shall be in accordance with MIL-STD-453.

4.3.2 Chemical analysis. Chemical analysis shall be made by wet chemical or spectrographic methods. In case of dispute, chemical analysis by wet chemical methods shall be the basis for acceptance.

4.3.3 Mechanical property tests.

4.3.3.1 Tensile strength. Tension tests shall be performed in accordance with requirements of method ASTM E-8, referenced in Fed. Test Method Std. No. 151.

4.3.3.2 Yield strength. The yield strength shall be determined by the offset or the extension under load method described in method ASTM E-8, referenced in Fed. Test Method Std. No. 151. In case of dispute, the offset method shall be used.

4.3.3.3 Workmanship. Each casting shall be carefully examined to determine conformance with this specification with respect to workmanship. Examination of a fraction of the inspection lot in accordance with a statistical sampling plan is acceptable provided the sampling plan is approved by the procuring activity.

4.3.4 Dimensions. Sufficient spot checks shall be made to ensure conformance with the dimensional tolerances specified on the applicable drawing.

4.4 Rejection and retests.

4.4.1 Rejection. If a test specimen or the castings fail to meet any of the tests required by this specification, the lot so represented shall be rejected.

4.4.2 Retests. Retests shall be permitted in accordance with the requirements of Fed. Test Method Std. No. 151.

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4.5 Inspection of preparation for delivery. The preservation, packaging, packing and marking shall be examined to determine conformance with the requirements of Section 5 of this specification, and as required by the specifications referenced herein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with MIL-STD-105, with inspection level S-3 and an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Unless otherwise specified (see 6.2), castings having finish machined surfaces shall be preserved and packaged in accordance with the applicable requirements of MIL-STD-649.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Unless otherwise specified (see 6.2), castings shall be packed in accordance with the applicable requirements of MIL-STD-649.

5.3 Marking.

5.3.1 Nonmilitary agencies. Unless otherwise specified in the contract or order, all shipping containers, bundles and unpackaged castings shall be marked with the name of the material, pattern, or part number as specified on the drawings, lot number, type, and composition of the castings, this specification number, the gross weight and quantity, the name of the contractor, and the number of the contract or order in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to the marking specified in 5.3.1 and special marking required in the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Alloys.

6.1.1.1 Alloy AZ63A is prone to excessive porosity and segregation with accompanying adverse effects on mechanical properties unless foundry procedures are closely controlled, and castings are limited to relatively simple shapes and sizes. Alloy AZ92A has slightly less resistance to corrosion than composition AZ63A. Alloy AM100A as a rule, is specified only when the parts to be cast are so intricate that the foundry would experience difficulty in using AZ92A.

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6.1.1.2 The effect of heat treating and aging on the corrosion resistance of the above castings is negligible, unless they are in contact with salt water. These castings find application in the F temper for lightly stressed parts; in the T4 temper they have high tensile strength, moderate yield strength and good elongation; in the T6 temper high tensile strength, high yield strength and low elongation. Castings in the T4 temper when subject to service condition temperatures of 200° F or higher, change rapidly to the T6 temper, accompanied by a small but measurable increase in dimensions. Increased stability of an order suitable for aircraft engines is attained by using the castings in the T5 or T6 temper.

6.1.1.3 Alloy AZ81A is supplied only in the T4 condition. It has superior casting characteristics combined with high tensile strength and good elongation. It was developed particularly for aircraft landing wheels. Alloy AZ91C is intended for use where maximum strength and hardness consistent with good ductility are needed. Alloy AZ92A has strength and hardness equal to or greater than AZ91C but has lower ductility.

6.1.1.4 The alloys AZ81A, AZ91C, and AZ92A are considered to have better foundry characteristics than AZ63A alloy.

6.1.1.5 Alloy EZ33A is intended for use where soundness and pressure tightness are required and where medium elevated temperatures, 300-500° F, are encountered. Alloys HK31A, HZ32A, and QE22A are used primarily for elevated temperature application up to 600° F.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Composition and temper of alloy (see 1.2.1 and 3.8).
(c) Protective treatment desired (see 3.10).
(d) Radiographic examination for soundness, if required, standards of acceptance percentage of castings to be inspected if other than 100 percent (see 3.3, 3.4, and 4.3.1).
(e) Location of critically stressed areas (see 3.9.3).
(f) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(g) Special markings, if applicable (see 5.3.2).
(h) Identification.

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